Chapter 1

Introduction to Finance and AI

1.1 Why AI in Finance

Finance has always been a quantitative discipline, but the questions it asks have outgrown the tools that were built to answer them. This part sets out what changed and how the field arrived here.

1.1.1 Why Finance Needs AI

Finance is one of the most data-rich and decision-intensive domains in the modern economy. Financial institutions collect vast amounts of information every day: prices, transactions, news, macroeconomic indicators, and firm-level reports. This information is often noisy, high-dimensional, and time-sensitive, which makes it hard to analyze with simple rules or intuition.

Financial markets are also highly competitive and dynamic, and firms, investors, and regulators must decide quickly and accurately. Methods from statistics, optimization, machine learning, and artificial intelligence have therefore become important in forecasting, risk management, trading, fraud detection, and customer service.

The central challenge in finance is not only to process data but to make decisions under uncertainty. A portfolio manager must decide how much risk to take. A bank must estimate the probability of default. A regulator must detect suspicious behavior before losses become large. In each of these cases, the decision-maker needs a model, a framework for judgment, and an understanding of the incentives and constraints of the market.

Figure 1.1 shows this process: raw data feeds a model, the model's output is checked before being trusted, and only then does it inform a decision. The book returns to that checking step when discussing why models must be interpreted rather than worshipped.

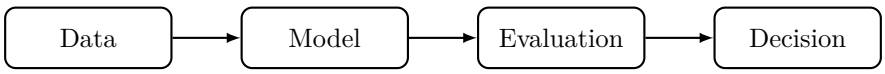


Figure 1.1: A simple pipeline from financial data to modeling, evaluation, and finally decisions

There is also an economic reason for AI's growing role. Many financial decisions are made with limited information, under time pressure, and in changing conditions, where fixed rules may miss subtle patterns in the data. Machine learning can uncover nonlinear relationships and adapt to new information, provided it is used with domain knowledge and careful validation.

Finance combines three layers of reasoning: economics, which supplies the structure of incentives; statistics, which estimates uncertain relationships from data; and implementation, which governs how a model behaves inside a real system with costs, delays, and operational constraints. A model that predicts a price movement becomes valuable only when it is linked to a trading rule, a risk limit, and the costs of execution. A credit

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model that predicts default must be read in light of borrower behavior, contract terms, and macroeconomic conditions. A strong practitioner is fluent in all three layers.

1.1.2 A Short History of Quantitative Methods and AI in Finance

Using mathematical models to make financial decisions is not new; what has changed over the past seventy years is the sophistication of the models and the scale of their data. A brief history shows why today's AI methods are the latest step in a long progression, not a sudden break.

The modern era of quantitative finance is often dated to 1952, when Harry Markowitz formalized portfolio selection as an optimization problem over expected return and variance (Markowitz 1952), work covered in depth in Chapter 12. This gave investors, for the first time, a rigorous mathematical language for the intuitive idea of diversification.

The next milestone came in 1973, when Fischer Black, Myron Scholes, and Robert Merton derived a closed-form formula for pricing options (Black and Scholes 1973; Merton 1973), the model introduced in Chapter 5. It turned derivatives trading from an informal, judgment-driven activity into something that could be systematically priced and hedged. Both developments used relatively simple, closed-form mathematics to capture an economically meaningful idea, and both remain in use today after decades of refinement.

The 1980s and 1990s saw the rise of algorithmic and electronic trading, as exchanges automated order matching and quantitative hedge funds began systematically exploiting statistical patterns in market data (Chapters 13 and 14). Around the same time, banks began applying statistical scoring models to credit decisions at scale, replacing purely judgment-based underwriting with models like those of Chapter 9.

Machine learning methods (decision trees, support vector machines, and eventually gradient boosting and random forests) entered mainstream financial use in the 2000s and 2010s as computing power and datasets grew, turning simple statistical scoring into much richer pattern recognition (Chapter 6).

Since roughly 2015, further developments have reshaped the field. Deep learning has made it practical to learn directly from large volumes of unstructured data (images, audio, and especially text), the foundation for the natural language processing methods in Chapter 16 and the alternative data methods in Chapter 17. More recently, large language models have been applied to summarizing earnings calls, extracting structured information from filings, and supporting research. Reinforcement learning has begun to be applied to sequential decisions such as trade execution, market making, and rebalancing, where an agent's actions today shape the state it faces tomorrow (Chapter 19). Agent-based modeling takes the opposite vantage point: it simulates many simple, interacting participants to see how bubbles, crashes, and contagion can emerge without any single agent intending them (Chapter 20). Table 1.1 summarizes this progression.

Period	Development
1950s	Markowitz mean-variance portfolio theory formalizes diversification mathematically
1970s	Black-Scholes-Merton option pricing enables systematic derivatives pricing and hedging
1980s–1990s	Electronic exchanges and algorithmic trading; statistical credit scoring at scale
2000s–2010s	Machine learning (trees, ensembles, support vector machines) applied to prediction and classification tasks across finance
2015–present	Deep learning for unstructured data; large language models applied to financial text; reinforcement learning for sequential decisions; agent-based models for market-wide simulation

Table 1.1: A brief timeline of quantitative methods in finance

The throughline is not that new methods replace old ones. Mean-variance optimization, Black-Scholes-Merton, and statistical scoring are all still in daily use, often alongside the newer machine learning methods. Each wave has instead expanded the range of problems that can be addressed and the data that can be used, while the economic questions (how to price risk, allocate capital, judge creditworthiness) have stayed largely

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the same. A new AI technique is generally a new way of answering an old financial question, not a reason to discard the economic reasoning behind it.

Table 1.1's AI/ML rows compress six decades of an older, independent field into two lines. AI research has its own timeline of breakthroughs and setbacks, most of it predating any financial application; Table 1.2 traces it.

Year	Development
1956	The Dartmouth Workshop, where John McCarthy and colleagues coin the term "artificial intelligence," is widely regarded as the field's founding event
1958	Frank Rosenblatt's Perceptron, the first trainable artificial neural network, generates enormous early excitement
1969–1986	Minsky and Papert's formal critique of the Perceptron's limitations helps trigger the first "AI winter", a prolonged pullback in AI funding and research interest
1986	Rumelhart, Hinton, and Williams popularize backpropagation, an efficient algorithm for training multi-layer neural networks (Chapter 6) that ends the first AI winter
1997	IBM's Deep Blue defeats reigning world chess champion Garry Kasparov, the first defeat of a sitting champion by a computer under tournament conditions
2012	AlexNet's decisive win in the ImageNet image-recognition competition demonstrates deep convolutional networks' advantage at scale, igniting the modern deep learning era (Chapter 17)
2016	DeepMind's AlphaGo defeats Go champion Lee Sedol using deep reinforcement learning (Chapter 19), a game long assumed too complex for a computer to master
2017	"Attention Is All You Need" (Vaswani et al. 2017) introduces the transformer architecture, the foundation of every large language model that follows (Chapter 16)
2020	OpenAI's GPT-3, a 175-billion-parameter language model, demonstrates that scale alone unlocks broad, few-shot task performance
2022	OpenAI's ChatGPT reaches mainstream adoption faster than any consumer software product before it, bringing generative AI into everyday use

Table 1.2: A timeline of key developments in AI and machine learning

Two patterns stand out against Table 1.2. First, progress was not steady: the field's first AI winter shows that a technique's early promise (the Perceptron) can collapse under a well-founded technical critique and take over a decade to recover, a caution worth remembering whenever a new method arrives with confident claims. Second, none of Table 1.2's milestones were built for finance: Deep Blue plays chess, AlexNet classifies photographs, AlphaGo plays Go. Each supplied a general-purpose capability (pattern recognition at scale, sequential decision-making, language understanding) that finance adopted only after it had been proven elsewhere. This is why Chapters 6, 16, 17, and 19 introduce these architectures and training algorithms as general-purpose machine learning before turning to what makes each useful in finance.

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1.2 The Financial Landscape

Before any model can be useful, the terrain it operates on has to be legible: who trades, what they trade, what data the activity leaves behind, and the handful of concepts every later chapter assumes. This part supplies that vocabulary.

1.2.1 A Brief Introduction to Financial Markets

Consider a mismatch found in every economy: a young firm has a profitable idea but no cash to build it, while a household nearby has idle savings. The household has no way to fund the idea directly, could not evaluate it if it could, and does not want to bear all the risk alone. Multiply this mismatch across millions of firms and households, with different amounts of capital, risk appetites, and time horizons, and the coordination problem becomes too large for buyers and sellers to solve by finding each other. A financial market solves this coordination problem at scale. It is a place where financial assets (stocks, bonds, currencies, commodities, and derivatives) are traded, letting individuals, businesses, and governments transfer capital, diversify risk, and price future uncertainty without every lender having to find and vet every borrower.

Financial markets do not exist only for speculation. They allocate resources across time and across economic agents: a household saving for retirement, a firm borrowing to expand, and a government issuing debt are all connected through markets with those who have capital to supply.

The value of an asset is determined by the expectations of market participants about future cash flows, risk, and liquidity. For example, the price of a stock reflects the market's assessment of a firm's future profitability, while the price of a bond reflects expectations about default risk and interest rates. Finance is therefore both empirical and forward-looking.

Some of the most important financial institutions include banks, which intermediate between savers and borrowers; asset managers, which invest funds on behalf of clients; insurance companies, which protect against specific risks; exchanges, which provide platforms for trading; and regulators, which oversee market integrity and stability. Each plays a distinct role in the financial system, and their incentives shape market behavior in subtle ways. Table 1.3 summarizes these roles.

Institution	Role
Banks	Intermediate between savers and borrowers
Asset managers	Invest funds on behalf of clients
Insurance companies	Protect against specific risks
Exchanges	Provide platforms for trading
Regulators	Oversee market integrity and stability

Table 1.3: Major financial institutions and their roles

Financial markets can be classified by what they trade. Equity markets trade ownership claims in firms. Bond markets trade debt securities. Money markets trade short-term borrowing and lending instruments, typically with maturities under one year (Chapter 3). Derivatives markets trade contracts whose value depends on other assets. Foreign exchange markets trade currencies. Each market has its own structure, participants, and incentives. Table 1.4 summarizes these market types.

Markets can also be described by the roles of their participants. Investors seek returns. Borrowers seek financing. Market makers provide liquidity by standing ready to buy and sell. Arbitrageurs exploit small pricing discrepancies. Regulators enforce rules and preserve confidence. These roles interact to create the price discovery process that makes markets informative. Table 1.5 summarizes these participant roles.

This perspective also explains why prices can move sharply on limited news. A market is not a single decision-maker but a collection of heterogeneous participants with different objectives, information, and horizons, so prices reflect a balance of competing forces rather than a simple average of known facts.

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Market	What is traded
Equity markets	Ownership claims in firms
Bond markets	Debt securities
Money markets	Short-term borrowing and lending instruments, such as Treasury bills, commercial paper, and repurchase agreements (Chapter 3)
Derivatives markets	Contracts whose value depends on other assets
Foreign exchange markets	Currencies

Table 1.4: Major types of financial markets

Participant	Objective or function
Investors	Seek returns
Borrowers	Seek financing
Market makers	Provide liquidity by standing ready to buy and sell
Arbitrageurs	Exploit small pricing discrepancies
Regulators	Enforce rules and preserve confidence

Table 1.5: Financial market participant roles

1.2.2 The Financial Data Ecosystem

Before turning to financial concepts, consider the data that feed AI applications in finance. Different data types require different tools and handling, and much of this book is organized around them. Table 1.6 summarizes the main categories.

Each data type has its own quality issues and conventions, and a recurring theme of this book is that using data well requires understanding where it comes from, not just how to load it into a model. Market data is timestamped and voluminous but expensive at high frequency and subject to microstructure noise (Chapter 2 introduces the handling of price and return data; Chapters 13 and 14 return to its highest-frequency form). Fundamental data is low-frequency and subject to reporting lag and accounting discretion (Chapter 4). Macroeconomic data is often revised after release, so a dataset built from final revised figures can overstate what was knowable in real time, a subtle look-ahead bias. Text and alternative data are the newest additions and, as Chapters 16 and 17 discuss, need far more preprocessing than the structured, tabular data of the earlier chapters.

A single financial AI system often combines several data types: a credit model may join a small business's fundamentals with its bank-account transactions, or a trading strategy may join market data with news sentiment. This is harder than it sounds. The data must be aligned in time and identity (the same firm, the same reporting period), and each source can fail or degrade independently.

1.2.3 Core Financial Concepts

A few central ideas in finance come before any AI application. They supply the vocabulary and intuition for financial problems, whether these are addressed with classical models or with machine learning.

Risk and Return

Suppose two investments are on offer: a government bond that reliably pays 3% a year, and a new company's stock that might return 30% or might lose half its value, with no way to know in advance which. Why would anyone hold the stock? Only because, on average and over time, it must offer more than the bond to compensate buyers for the chance of a large loss. This is the risk-return trade-off, a fundamental principle of finance: higher expected return usually comes with higher risk, because otherwise investors would hold the safer asset. Every investor must decide how much uncertainty to bear in exchange for potential reward, a trade-off central to portfolio choice, asset pricing, and corporate finance.

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Data type	Description and examples
Market data	Prices, quotes, trading volume, and order-book data, ranging from daily closing prices to microsecond-level tick data (Chapters 3, 13, 14)
Fundamental data	Accounting and financial statement data reported by firms, such as revenue, earnings, and balance sheet items (Chapter 4)
Macroeconomic data	Interest rates, inflation, unemployment, and other economy-wide indicators that affect asset prices broadly (Chapter 7)
Credit and transaction data	Loan repayment history, credit bureau data, and payment transaction records used in credit and fraud models (Chapters 9, 15)
Operational loss and incident data	Internal and external operational-loss event records, incident and audit narratives, and system and access logs used to model and detect operational failures (Chapter 10)
Insurance policy and claims data	Policy, exposure, claims, telematics, and catastrophe-model data used to price, reserve, and transfer insured risk (Chapter 11)
Text and news data	Earnings call transcripts, news articles, analyst reports, and social media, used for sentiment and information extraction (Chapter 16)
Alternative data	Non-traditional sources such as satellite imagery, web traffic, or shipping data that proxy for economic activity (Chapter 17)

Table 1.6: Major categories of financial data

Figure 1.2 illustrates the idea: higher expected returns are typically obtained only by accepting greater variability in outcomes.

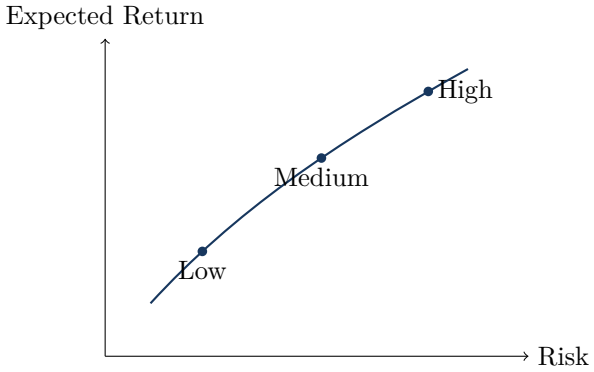


Figure 1.2: The risk-return trade-off: higher expected returns generally require accepting greater risk

Time Value of Money

Suppose someone offers to pay a debt either with $1,000 today or $1,000 in five years. Even setting aside inflation and the risk that the payer defaults, the two offers are not equivalent, and a quantitative reader's first instinct that a dollar is a dollar needs correcting: $1,000 today can be invested and grow into more than $1,000 by the time five years have passed, so accepting the delayed payment means giving up that growth

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for nothing. Money available today is worth more than the same nominal amount in the future because it can earn a return in the meantime. This principle underlies discounting, present value, interest rates, and valuation: future cash flows must be converted into present value before they can be compared fairly, a step central to bond pricing, project valuation, and investment analysis.

The future value of an amount PV invested today at rate r for n years is

$$FV = PV(1 + r)^n,$$

and Table 1.7 applies this formula to \$1,000 invested at three different rates over three different horizons.

Annual rate	10 years	20 years	30 years
3%	\$1,343.92	\$1,806.11	\$2,427.26
6%	\$1,790.85	\$3,207.14	\$5,743.49
9%	\$2,367.36	\$5,604.41	\$13,267.68

Table 1.7: Future value of \$1,000 invested at different rates and horizons

Two patterns in Table 1.7 recur throughout the book. First, compounding is highly nonlinear in the rate: doubling the rate from 3% to 6% more than doubles the 30-year future value, and tripling it to 9% multiplies the 30-year value by more than five. Second, the horizon matters enormously: at 9%, the difference between a 10-year and a 30-year horizon is a factor of roughly 5.6, though the annual rate never changed. Present value, used constantly from Chapter 5 onward, is this same relationship read in reverse:

$$PV = \frac{FV}{(1 + r)^n},$$

discounting a future amount back to today.

Diversification

Suppose an investor holds shares in a single airline. A jump in fuel prices, a pilots' strike, or a crash investigation can each wipe out a large fraction of the stock's value, and all three risks are specific to that company and its industry. Now suppose the investor splits the same money across an airline, a software company, and a grocery chain, three businesses with largely unrelated day-to-day fortunes. A fuel-price spike still hurts the airline but does little to the other two, so news that would devastate an all-airline portfolio barely dents the diversified one. This is diversification: spreading investments across assets or risks so that no single loss can sink the portfolio. The intuition is simple (not all investments move together), but the consequence is not: diversification is one of the few strategies in finance that reduces risk without necessarily reducing expected return, which is why Chapter 12 treats it mathematically rather than as a rule of thumb. In portfolio construction, diversification reduces idiosyncratic risk, the risk specific to one company or industry, while retaining exposure to the broad economic factors that no amount of spreading across stocks can eliminate.

Market Efficiency

Suppose a quantitative analyst backtests a simple rule (buy a stock the day after it reports earnings above analyst expectations) and finds it would have produced a solid excess return over the past decade. Should the analyst expect it to keep working once real money is committed? That depends on how much of the information was already in the price before the analyst could act: if thousands of other participants saw the same earnings beat and traded within minutes, little may be left by the next morning.

The efficient market hypothesis formalizes this concern: asset prices reflect available information to a large extent. This has important consequences for forecasting, trading, and the interpretation of abnormal returns. Markets are not perfectly efficient at all times, but the framework remains a core reference point.

For quantitative readers, this motivates the distinction between signal and noise. Market efficiency does not say that no one can beat the market. It says the relevant question is often not whether a signal exists, but whether it survives competition from other informed participants, delays in data availability, transaction costs, model risk, and market frictions.

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1.3 What This Book Covers

With the landscape in place, the practical question is which financial problems machine learning improves on, and in what order they are best learned. This part maps the problems onto the chapters that treat them and fixes the notation; the preface explains the chapter order.

1.3.1 Financial Problems That AI Can Help Solve

AI is useful in finance because many important problems involve prediction, classification, optimization, and decision-making under uncertainty: forecasting asset prices or returns, predicting credit default or bankruptcy, detecting fraudulent transactions, monitoring for operational failures and misconduct, optimizing portfolios, identifying trading opportunities, and extracting information from financial text and news. None of these is a generic data problem; the underlying financial model, market structure, and regulatory context matter. A credit model must reflect the economics of default, and a trading strategy must account for transaction costs, market liquidity, and execution risk.

This is a key point for readers with a machine learning background. A model that performs well on a generic prediction task may fail in finance if it ignores institutional details: a pricing model must respect no-arbitrage conditions, and a forecasting model that ignores transaction costs, liquidity constraints, or regime changes may perform well in sample and still fail in practice. Financial data are also generated by strategic behavior: when a firm announces earnings or a central bank changes interest rates, the data record the reactions of many agents as well as economic fundamentals. Finance is a structured domain with economic logic and constraints, and it demands both predictive accuracy and economic plausibility.

Table 1.8 organizes the main problem types by machine learning task and points to where each is covered, previewing the book's structure problem-first rather than chapter-first.

Three things stand out from this taxonomy. First, several finance problems map onto more than one task type. Portfolio construction is an optimization problem but often uses regression or dimensionality reduction as an input, which is why later chapters combine methods from Chapter 6 with domain-specific structure. Second, some task types in Table 1.8 recur across many chapters (classification appears in credit, trading, and fraud) because the statistical machinery, introduced once in Chapter 6, is reused rather than re-derived for each application. Third, the table's last row is an outlier: every other row asks a model to predict, classify, or optimize for a single decision-maker, while multi-agent simulation asks what emerges when many simple decision-makers interact (Chapter 20).

1.3.2 Notation and Conventions Used in This Book

A few notational conventions are used throughout the book. Random variables and their realizations are both written in italics (for example, X or x), with the distinction clear from context. A hat denotes an estimate: $\hat{\beta}$ is an estimated coefficient and $\hat{y}$ a predicted value, while β and y denote the true, generally unobserved, parameter and outcome. Vectors and matrices are written in bold only where the distinction from scalars is not otherwise clear from context, and $\|\cdot\|$ denotes the Euclidean norm. Time subscripts are written as X_t, with $t-1$ denoting the immediately preceding period, and

$$\Delta X_t = X_t - X_{t-1}$$

denoting a first difference.

Monetary amounts are given in US dollars unless stated otherwise, and percentages are used interchangeably with decimals (5% and 0.05 mean the same thing) depending on which is clearer in context. Interest rates, returns, and growth rates are annualized unless a different frequency (monthly, daily) is explicitly stated. Every worked numerical example can be checked against the chapter's companion notebook, as the preface explains.

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Task type	Example financial problem	Covered in
Regression	Forecasting returns, volatility, an insurance claim cost, or a credit spread	Ch. 6, 7, 11
Classification	Predicting default, fraud, an operational event type, a claim's severity tier, or a trading signal's direction	Ch. 6, 9, 10, 11, 15
Time series forecasting	Forecasting interest rates, volatility, or macro indicators	Ch. 7
Distribution and tail estimation	Estimating a loss distribution and its far-tail quantiles: market VaR, operational-loss capital, insurance pricing and solvency	Ch. 8, 10, 11
Optimization	Constructing a portfolio subject to risk and constraint limits	Ch. 12
Clustering / dimensionality reduction	Grouping similar firms or compressing a large factor set into a few drivers	Ch. 6, 7, 12, 17
Sequential decision-making	Executing a trade over time to minimize impact, formalized as a Markov decision process	Ch. 13, 14, 19
Natural language processing	Extracting sentiment or information from text	Ch. 16
Anomaly detection	Flagging unusual transactions, account behavior, operational near-misses, or implausible insurance claims and coding patterns	Ch. 10, 11, 15, 17
Explainability and fairness auditing	Explaining an individual model decision and testing a model for disparate impact	Ch. 18, 21
Multi-agent simulation	Modeling how many simple, interacting participants jointly produce market-wide bubbles, crashes, or contagion	Ch. 20

Table 1.8: A taxonomy of AI problem types in finance

1.4 Reading Finance Beyond the Numbers

A financial statement is a set of numbers, but it is also a set of claims made by people with incentives. This part introduces the accounting that later chapters model, and the reasons a purely numerical reading of it will mislead.

1.4.1 Financial Statements, Cash Flows, and Valuation

Although much of modern finance is expressed through prices, rates, and stochastic models, the underlying economic reality is simpler: firms create value by generating cash flows over time. A company can be profitable on paper and still face financial distress if it does not generate enough cash to meet obligations. Financial analysis therefore begins with a careful reading of the balance sheet, the income statement, and the cash flow statement.

The balance sheet summarizes what a firm owns and what it owes at a point in time. Assets include cash, inventories, property, and intellectual property. Liabilities include debt, payables, and other obligations. Equity represents the residual claim of owners after liabilities are accounted for. The income statement shows revenues, expenses, and profit over a period of time. The cash flow statement, by contrast, explains how cash moved through the business and is often more informative than accounting earnings when one is trying to understand solvency and future financing needs.

The distinction matters because accounting profits and economic value differ. A firm may report large earnings while consuming cash through inventory accumulation, capital expenditure, or working capital needs. Investors and lenders ask not only whether the firm appears profitable, but whether it can sustain operations

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and meet future obligations. A predictive model may forecast earnings, but an analyst must also ask whether those earnings are backed by cash generation and durable competitive advantages.

Valuation rests on this logic. The value of an asset is the present value of the expected future cash flows that it will generate, discounted at an appropriate rate. In symbols, a simple present value relation is

$$V = \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t}, \tag{1.1}$$

where CF_t denotes the cash flow in period t and r is the discount rate. This expression is one of the most important in finance because it unifies bond pricing, stock valuation, project appraisal, and much of corporate analysis. The challenge is that future cash flows are uncertain and the discount rate must reflect both time and risk, which is where statistics and econometrics become essential.

Valuation is rarely mechanical. A company with strong historical performance may still lose value if its market position erodes or interest rates rise sharply. A machine learning model that forecasts revenues or defaults accurately should still have its output checked against the firm's accounting structure, debt obligations, industry, and management incentives. The best systems combine statistical learning with financial reasoning and careful validation.

1.4.2 Why Finance Is Not Just a Collection of Numbers

A common misconception is that finance is the manipulation of numerical data. Finance is about incentives, contracts, and the allocation of resources under uncertainty, and the numbers in financial statements or market data are outcomes of institutional processes: decisions about borrowing, investing, pricing, regulation, and risk management. A quantitative analyst should therefore not be satisfied with a high-performing prediction model without a clear economic explanation for why it works. A pattern in stock returns may reflect risk exposure, information diffusion, transaction costs, or temporary market inefficiency. A default model may be capturing leverage, deteriorating cash flow, or sector-level stress. The signal is meaningful only when interpreted through finance.

This is also why financial modeling must be careful about causality. Correlation is not enough. A variable may be associated with future returns or defaults for reasons that do not persist once conditions change, so a sensible practitioner tests assumptions, monitors model stability, and judges whether a discovered pattern is economically plausible.

1.4.3 A Practical Perspective on the Role of AI in Finance

The purpose of AI in finance is to augment financial expertise, not to replace it. AI methods can process large and complex data, detect nonlinear patterns, and automate tasks too time-consuming for humans. Their usefulness depends on the problem formulation: a poorly defined objective yields a model that looks sophisticated but makes poor decisions. A successful practitioner translates a financial question into a statistical problem, and then the model's output back into a financial decision.

Consider the example of credit scoring. A bank may want to estimate the probability that a borrower will default over the next year. A machine learning model can use many variables, including repayment history, income, occupation, and account activity. Yet the model is most useful when it is tied to domain knowledge about loan structure, collateral, legal obligations, and the macroeconomic environment. Otherwise, it may learn patterns that reflect temporary conditions rather than durable credit risk.

This view also has organizational consequences that a purely technical treatment overlooks. Deploying an AI system in a financial institution typically requires sign-off from risk management, compliance, and sometimes a model validation team whose job is to challenge the model before it is used with real capital or customers, a process called model risk management. A model that cannot be explained to these stakeholders is often not deployable at all, whatever its statistical performance, which is part of why Chapter 18 treats explainability as a first-class technical requirement. This institutional layer can feel slow to a reader from machine learning, but a financial model's errors are rarely private. A mispriced loan, a bad trade, or a missed fraud case can affect customers, counterparties, and, in aggregate, the stability of the financial system, which is why financial AI is regulated more heavily than AI that recommends movies.

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1.4.4 Finance as a System of Incentives and Information

Section 1.4.2 made this point for a single model: a statistical pattern is meaningful only once it has an economic explanation. This section takes the idea up to the level of a whole market, where the incentives shaping how data are generated deserve naming directly. A manager may be rewarded for short-term earnings growth even when that growth is not sustainable. A lender may extend credit to a borrower who appears safe in normal conditions but becomes fragile when conditions deteriorate. A regulator may face pressure to support market confidence even when this increases moral hazard. Finance is intertwined with human behavior and institutional design.

Financial data are therefore records of strategic interaction, not passive observations. Prices aggregate the small pieces of information dispersed across many participants, which makes price data valuable, but they also encode noise, strategic behavior, and market frictions. The analyst must separate signal from noise while remembering that the data are generated by participants with incentives.

Two well-documented episodes show how badly things can go wrong when a model ignores the incentives shaping its data. Before the 2008 financial crisis, statistical models of mortgage default risk were built on data from a period of rising home prices and relatively loose underwriting. They implicitly assumed that the incentives generating that data would persist, in particular that mortgage originators bore meaningful consequences for making bad loans. The rise of mortgage securitization changed those incentives: many originators sold loans onward quickly enough to bear little of the eventual default risk, so underwriting quality deteriorated in ways the historical models never observed. The models were not statistically wrong in a narrow sense; they were estimated correctly on the data available. They failed because the economic process generating the data had shifted underneath them, the non-stationarity concern developed further in Chapter 9.

A second, smaller-scale illustration is Goodhart's Law, often summarized as "when a measure becomes a target, it ceases to be a good measure." A credit score starts as a statistical summary that correlates with creditworthiness. Once borrowers and lenders know that a numerical threshold controls loan approval or pricing, some participants will alter their behavior to cross it, whether or not their creditworthiness has changed. This is the reflexivity theme introduced in Chapter 6: unlike a model of a physical system, a financial model's targets can adapt to the model itself, one more reason financial AI requires ongoing monitoring rather than one-time validation.

1.5 Reasoning Under Uncertainty

Every financial decision is a bet made with incomplete information, which makes probability the native language of the subject. This part develops how to size such a bet, what new information is worth, and why a model's output is an argument rather than an answer.

1.5.1 Uncertainty, Probability, and Decision-Making

Finance is fundamentally about decisions under uncertainty. A lender does not know with certainty whether a borrower will repay. An investor does not know whether an asset price will rise or fall. A firm does not know whether a new project will be profitable. The question is not whether uncertainty can be eliminated, because it usually cannot, but how decision-makers should act when outcomes are uncertain and the costs of being wrong are not symmetric.

Finance runs into two different kinds of not-knowing. The economist Frank Knight drew the distinction in 1921. *Risk* describes a situation where the possible outcomes are known and their probabilities can be estimated, whether from a well-understood physical process (a fair coin) or from a long, stable history of similar events. *Uncertainty* (sometimes called Knightian uncertainty) describes a situation where even the set of possible outcomes, let alone their probabilities, cannot be fully specified in advance. A casino game is risk in this sense; a novel financial crisis, a first-of-its-kind fraud scheme, or a new technology's effect on an entire industry is uncertainty. Nearly every probabilistic model in this book (the Value at Risk models of Chapter 8, the credit-scoring models of Chapter 9, the volatility forecasts of Chapter 7) implicitly treats its subject as risk, assuming that a historical sample is enough to pin down the relevant probabilities. Before

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trusting a model's probability estimate, ask whether the situation is closer to a repeatable game with a stable, estimable distribution, or to a novel one where the model's confidence may be misplaced.

This is where probability and expected value become central. If an investor faces two possible outcomes, a gain and a loss, the expected value of the gamble may be positive even if the investor would prefer not to take the risk. This motivates the idea of risk aversion. A risk-averse decision-maker usually requires a larger expected payoff to accept a gamble with greater uncertainty, a logic that underlies portfolio choice, corporate financing, and insurance.

A simple expression for expected value is

$$\mathbb{E}[X] = \sum_i p_i x_i, \tag{1.2}$$

where p_i is the probability of state i and x_i is the associated payoff. In finance these probabilities are rarely known precisely and can change, so financial models must be updated as new information arrives.

A small numerical example makes the idea of risk aversion concrete. Suppose an investor is offered a gamble with a 50% chance of gaining \$20,000 and a 50% chance of losing \$10,000. The expected value of the gamble is

$$\mathbb{E}[X] = 0.5(20{,}000) + 0.5(-10{,}000) = \$5{,}000,$$

which is positive, so a risk-neutral decision-maker, one who cares only about expected value, should accept it. Yet many investors would decline, because the possibility of losing \$10,000 matters more to them than the expected value suggests. Economists capture this with a concave utility function $u(\cdot)$: the investor evaluates the gamble by its expected utility, $\mathbb{E}[u(X)]$, rather than its expected value. The certainty equivalent is the guaranteed amount that gives the same expected utility as the gamble. For a risk-averse investor it is always less than the \$5,000 expected value, and the gap, sometimes called the risk premium, measures how much the investor would pay to avoid uncertainty.

Making this concrete requires a specific utility function, applied to the investor's total wealth rather than the gamble's payoff alone, since a concave function is meaningful only at actual wealth levels. Suppose the investor's wealth today is $W_0 = \$50{,}000$, and take the square-root utility function $u(W) = \sqrt{W}$, a standard textbook example of a risk-averse (concave) utility function. The gamble leaves the investor with wealth $W_0 + 20{,}000 = \$70{,}000$ or $W_0 - 10{,}000 = \$40{,}000$, so

$$\mathbb{E}[u(W)] = 0.5\sqrt{70{,}000} + 0.5\sqrt{40{,}000} \approx 0.5(264.575) + 0.5(200.000) = 232.288.$$

The certainty equivalent wealth CE solves $u(CE) = \mathbb{E}[u(W)]$, that is, $\sqrt{CE} = 232.288$, giving $CE \approx \$53{,}957.51$, noticeably less than the expected wealth of $0.5(70{,}000) + 0.5(40{,}000) = \$55{,}000$. The risk premium is the difference,

$$\text{Risk premium} = \$55{,}000 - \$53{,}957.51 \approx \$1{,}042.49,$$

which has a direct interpretation: this investor would accept a guaranteed \$53,957.51 in place of the gamble, paying up to \$1,042.49 to avoid its uncertainty even though its expected wealth is higher. The same comparison of an uncertain payoff with its certainty equivalent underlies the pricing of insurance, the equity risk premium, and the risk-return trade-off introduced earlier. It reappears in Chapter 12's mean-variance framework as the reason a risk-averse portfolio manager accepts a lower expected return in exchange for lower volatility.

1.5.2 The Kelly Criterion: How Much to Bet, Not Just Whether

A positive expected value answers whether a favorable gamble is worth taking at all. A repeatable opportunity, the same favorable bet placed over and over, raises a sharper question: what *fraction* of current wealth should be wagered each time? Betting too little leaves edge on the table; betting too much, as this section shows, can make an investor with a real, positive edge poorer with certainty over time.

The question is nontrivial, rather than "bet as much as a positive edge allows," because wealth wagered repeatedly compounds rather than accumulates. A single bet's expected value is an arithmetic average across outcomes, the right tool for deciding whether to take that bet once. Wealth carried across many repetitions is a product of per-round multipliers, $(1 + f)$ after a win and $(1 - f)$ after a loss, and a quantity built from

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repeated multiplication is governed by its *geometric* mean. A bet size can raise the arithmetic average of outcomes while driving the geometric average, and with it long-run wealth, toward zero; this asymmetry, not risk aversion in the subjective sense discussed above, is what makes betting too large a fraction ruinous.

The Kelly criterion answers this for a repeatable bet with probability p of winning (paying b dollars per dollar wagered) and probability $q = 1 - p$ of losing the full amount wagered: the fraction of wealth that maximizes the long-run *geometric* growth rate of wealth is

$$f^* = p - \frac{q}{b}. \tag{1.3}$$

For an even-money bet ($b = 1$) with a 60% win probability, $f^* = 0.6 - 0.4 = 0.2$: wager 20% of current wealth on each repetition, no more and no less. The formula is quoted here rather than derived. The book's companion technical note on the Kelly criterion works the argument through in full: why f^* is a strict global maximum rather than merely a stationary point, why maximizing growth is equivalent to maximizing expected logarithmic wealth, and why the optimal fraction can rationally exceed one, recommending leverage, once a loss costs less than the entire stake.

Why "no more" matters as much as "no less" is the surprising part. The long-run growth rate of wealth from repeatedly wagering a fraction f is $g(f) = p\ln(1+f) + q\ln(1-f)$; evaluating it at three candidate fractions for this same 60%-favorable bet,

$$g(0.2) \approx 0.0201, \qquad g(0.5) \approx -0.0340, \qquad g(1.0) = -\infty,$$

shows that wagering half of wealth on every repetition, although each bet still has positive expected value, produces a *negative* long-run growth rate. An investor following this rule is mathematically certain to see wealth decay toward zero over enough repetitions, not merely higher volatility around a positive trend.

Wagering everything each time ($f = 1$) is worse still, guaranteed eventual ruin, since a single loss wipes out the entire stake. A decision-maker who evaluates only a single bet's expected value, without asking how large a fraction of wealth to commit when it repeats, can be led to a sizing decision that is ruinous. This position-sizing discipline resurfaces in Chapter 12's discussion of leverage and Chapter 13's execution-sizing decisions.

1.5.3 The Value of Information: What Is a Perfect Signal Worth?

A closely related question asks not how much to bet, but how much a decision-maker should be willing to pay for information that resolves uncertainty *before* a decision is made. Suppose a firm is deciding whether to fund a $2 million project, an idealized version of the investment decisions that Chapter 5's discounted-cash-flow and real-options material treats in detail. The project succeeds with probability 0.4, returning a net gain of $5 million, or fails with probability 0.6, losing the full $2 million invested. Without any additional information, the expected value of proceeding is

$$\mathbb{E}[\text{Proceed}] = 0.4(5) + 0.6(-2) = \$0.8 \text{ million},$$

positive, and better than the $0 from not proceeding at all, so a rational, risk-neutral firm proceeds, accepting the 60% chance of a $2 million loss along with the 40% chance of a $5 million gain.

Now suppose the firm could instead consult a perfectly accurate forecaster, at some price, who reveals with certainty which state (success or failure) will occur before the firm must commit. Knowing this, the firm would proceed only in the success state and decline in the failure state, earning

$$\mathbb{E}[\text{Proceed} \mid \text{perfect information}] = 0.4(5) + 0.6(0) = \$2.0 \text{ million},$$

since the $2 million loss is avoided entirely in the 60% of cases where failure would otherwise have occurred. The expected value of perfect information (EVPI) is the difference between these two figures,

$$\text{EVPI} = 2.0 - 0.8 = \$1.2 \text{ million},$$

the maximum a rational firm should pay for a perfectly accurate signal before deciding, since paying more would make consulting the forecaster worse than proceeding blind. No real forecast, credit model, or market signal is perfectly accurate, so EVPI is an upper bound. It answers "how much could better information possibly be worth here," a question worth asking before paying for additional data, a more sophisticated model, or a costly due-diligence process.

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1.5.4 Bayesian Updating: Learning from New Information

A related idea, which underlies how many machine learning systems in this book are best interpreted, is that beliefs about an uncertain quantity should be updated, not replaced, as new evidence arrives. Bayes' theorem formalizes this:

$$\Pr(A \mid B) = \frac{\Pr(B \mid A)\ \Pr(A)}{\Pr(B)}, \tag{1.4}$$

where $\Pr(A)$ is the prior probability of some hypothesis A before seeing evidence B, $\Pr(B \mid A)$ is the likelihood of observing that evidence if A is true, and $\Pr(A \mid B)$ is the posterior probability of A after accounting for the evidence.

An example from fraud detection (Chapter 15) shows why this matters. Suppose 1% of transactions at a bank are fraudulent, so $\Pr(\text{Fraud}) = 0.01$. A fraud detection model correctly flags 95% of actual fraud, $\Pr(\text{Flag} \mid \text{Fraud}) = 0.95$, but also flags 3% of legitimate transactions, $\Pr(\text{Flag} \mid \text{Not Fraud}) = 0.03$. Given that a transaction has been flagged, what is the probability it is actually fraudulent? Applying Bayes' theorem,

$$\begin{aligned}
\Pr(\text{Flag}) &= \Pr(\text{Flag} \mid \text{Fraud})\Pr(\text{Fraud}) + \Pr(\text{Flag} \mid \text{Not Fraud})\Pr(\text{Not Fraud}) \\
&= 0.95(0.01) + 0.03(0.99) \\
&= 0.0392, \\
\Pr(\text{Fraud} \mid \text{Flag}) &= \frac{0.95(0.01)}{0.0392} \approx 0.242.
\end{aligned}$$

Despite a model that sounds highly accurate (95% detection, only 3% false alarms), a flagged transaction is fraudulent only about 24% of the time. Fraud is so rare that even a small false-positive rate produces many more false alarms than true detections in absolute terms. The same base-rate intuition drives the discussion of imbalanced classification in Chapter 6, and the example shows quantitatively why accuracy alone is a poor way to evaluate a model applied to a rare event.

Bayesian updating also describes how a well-run financial model should be maintained: rather than fixing a model's parameters forever, or discarding the model the first time it errs, each new batch of data revises a belief incrementally, and today's posterior becomes tomorrow's prior. This is the spirit behind the ongoing model monitoring of Chapter 18 and the state-space and Kalman-filtering methods of Chapter 7, which are Bayesian updating carried out recursively over time.

The "posterior becomes the next prior" idea is easiest to see by continuing the fraud example. Suppose the flagged transaction is now compared against the customer's typical spending location, and it occurred somewhere the customer has never transacted before, a pattern found in 60% of fraud cases but only 10% of legitimate ones: $\Pr(\text{Unusual Location} \mid \text{Fraud}) = 0.6$ and $\Pr(\text{Unusual Location} \mid \text{Not Fraud}) = 0.1$. Rather than starting over, a Bayesian analysis treats the first posterior, $\Pr(\text{Fraud} \mid \text{Flag}) \approx 0.242$, as the prior for this second update:

$$\begin{aligned}
\Pr(\text{Location}) &= 0.6(0.242) + 0.1(1 - 0.242) = 0.221, \\
\Pr(\text{Fraud} \mid \text{Flag}, \text{Location}) &= \frac{0.6(0.242)}{0.221} \approx 0.657.
\end{aligned}$$

Two independent pieces of weak-to-moderate evidence, an automated flag and an unusual location, combine to raise the probability of fraud from a 1% base rate to roughly 66%, each update using the same formula with the previous posterior in place of the original prior. Figure 1.3 lays out the chain. A production fraud system (Chapter 15) accumulates evidence from several model outputs in this way before deciding whether a transaction warrants manual review, which is why several weak signals, evaluated one Bayesian update at a time, can be more reliable than any single signal.

The step that makes this chain legitimate is easy to miss. Strictly, the second update needs the likelihood of an unusual location given fraud *and* given that the transaction was already flagged, $\Pr(\text{Location} \mid \text{Fraud}, \text{Flag})$, not $\Pr(\text{Location} \mid \text{Fraud})$. Using the latter assumes that the two signals are *conditionally independent*: once the true state, fraud or not, is known, learning that the flag fired says nothing more about the location. That is rarely true. A flagging model that already leans on location-type features will tend to flag legitimate transactions made while the customer is traveling, so among flagged legitimate transactions an unusual location is more common than the 10% seen among legitimate transactions overall. If it is 30% instead, with

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the fraud likelihood unchanged at 60%, the second update gives $0.6(0.242)/[0.6(0.242) + 0.3(0.758)] \approx 0.390$, about 39% rather than 66%; and if the location signal were entirely redundant given the flag, the posterior would stay at 24%. Treating correlated signals as independent counts the same evidence twice and makes the system overconfident, which is the assumption, and the failure mode, of the Naive Bayes classifier in Chapter 6.

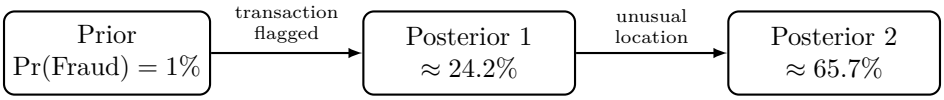


Figure 1.3: The fraud example's two-step Bayesian update chain

1.5.5 Why Models Must Be Interpreted, Not Worshipped

A common temptation in finance is to trust a model because it is sophisticated or produces attractive historical results. A model can be mathematically elegant and still economically misleading: a credit model may appear accurate in a stable period and become unstable during a recession, and a fraud model may detect obvious anomalies but miss subtle forms of misconduct that change over time. The evaluation step in Figure 1.1's pipeline is not a formality; it is where an overconfident model gets caught, or does not. Model evaluation in finance is therefore an economic and operational exercise as well as a statistical one. It asks not only whether the model fits the data but whether its assumptions make sense, its outputs are usable, and the implementation is robust.

The 1998 collapse of Long-Term Capital Management (LTCM) is a case study in this failure mode, examined further in Chapter 3. The hedge fund's partners included two Nobel laureates in economics, and its models were, by the standards of the time, extremely sophisticated. They correctly estimated how fixed-income spreads behaved under historically normal conditions, but had no mechanism for anticipating a regime in which previously uncorrelated markets, including Russian sovereign debt, moved together in a global flight to quality. At that same moment the fund's highly leveraged positions made it a forced seller into a market with too few buyers. No model in later chapters, however sophisticated, is exempt from this risk: a model is only as good as the range of conditions it was built and tested to anticipate. That is why the stress-testing and scenario-analysis techniques of Chapter 8 are a check on quantitative models generally, not a technique reserved for portfolio risk.

1.6 Limits, Challenges, and Practice

Nothing developed so far works as cleanly in production as it does on a page. This part collects the obstacles that recur throughout the book, and closes with a case in which classical finance and machine learning were combined successfully.

1.6.1 Challenges of Applying AI in Finance

Several challenges recur whenever AI methods are applied to financial problems; later chapters revisit each in depth.

Data challenges. Financial data are noisy, unevenly sampled, subject to reporting delays, and sometimes revised after the fact. Historical datasets can also suffer from survivorship bias, in which failed firms or discontinued strategies are quietly excluded, making the past look more favorable than it was.

Non-stationarity. Unlike many physical systems, financial markets change their own statistical behavior over time, in response to new regulation, new technology, changing investor composition, and the actions of other participants, including other AI systems. A model calibrated on one period can fail in the next without any error in how it was built.

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Overfitting and backtest illusions. Because financial data are limited relative to the number of strategies and features that can be tried, it is easy to find a pattern that worked in history by chance. A strategy that looks excellent in a backtest can perform poorly, or lose money, once deployed with real capital.

Interpretability and accountability. Many financial decisions, particularly credit and insurance decisions, are subject to legal requirements that they be explainable and non-discriminatory. This constrains how opaque a model can be, whatever its statistical performance (Chapter 18).

Operational and execution frictions. A model's statistical output is not a financial outcome. Transaction costs, market impact, latency, and operational failures can erode or eliminate a theoretical edge, which is why this book emphasizes economic as well as statistical evaluation.

Ethical and systemic considerations. Widely adopted AI models can create correlated behavior across many market participants, potentially amplifying booms and busts. Models used in lending, insurance, or hiring can also encode or amplify existing biases in historical data if they are not carefully audited.

These are not reasons to avoid AI in finance but to apply it carefully, with an understanding of financial institutions, incentives, and market structure alongside statistical skill, the combination this book aims to build.

1.6.2 Case Vignette: How Classical Finance and AI Combine in Practice

To make the chapter's themes concrete, consider a stylized (fictional) example that previews how the pieces of this book fit together. A mid-sized asset manager wants to build a systematic equity strategy that selects stocks likely to outperform over the coming quarter.

The manager does not start with machine learning. The process begins with the financial statement analysis of Chapter 4: screening a universe of firms on profitability, leverage, and cash generation, and discarding firms with clear signs of distress before any prediction model is involved. A purely data-driven model applied to raw price history would not make this economic judgment on its own.

Next, the manager builds a predictive model with the regression and classification tools of Chapter 6 to rank the remaining firms by expected relative performance. Its features draw on several data types from Table 1.6: fundamental ratios from the screening step, momentum and volatility measures from the time series methods of Chapter 7, and, increasingly, sentiment scores extracted from earnings-call transcripts with the natural language processing methods of Chapter 16. The model is evaluated with walk-forward validation, not a random train-test split, since a random split would let the model see information from the future relative to some of its training data.

The ranked predictions are an input to portfolio construction, not the final answer. Chapter 12's methods convert a ranking into position sizes that respect diversification, risk limits, and transaction costs, since the highest-ranked stock might be too risky, too illiquid, or too correlated with the rest to hold at a large weight. Finally, the execution methods of Chapter 13 determine how to buy and sell the positions without moving prices adversely, and the risk models of Chapter 8 monitor whether the realized portfolio behaves as expected.

At every step, a classical financial concept (distress screening, diversification, transaction costs) sits alongside a modern statistical or machine learning method, and neither is sufficient on its own. This interleaving, not the replacement of one by the other, is the organizing principle of every applied chapter that follows.

1.7 Key Takeaways

Finance is a domain in which data, uncertainty, and decision-making interact closely. Quantitative readers can progress quickly by connecting mathematical tools with financial intuition, but they must learn to interpret models in light of market structure, incentives, and institutional constraints. The most effective applications of AI in finance combine statistical skill, economic understanding, and practical judgment.

The history in Section 1.1.2 showed that today's machine learning methods extend, rather than replace, decades of classical quantitative finance, and the data ecosystem in Section 1.2.2 previewed the data types later chapters draw on, each with its own quality issues. The numerical examples (compounding in Table 1.7, the gamble and its certainty equivalent, the fraud-detection application of Bayes' theorem) each illustrate a principle that reappears in more technical form later: nonlinear compounding, risk aversion, and the danger

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of ignoring base rates and of double-counting correlated evidence. Finally, the case vignette in Section 1.6.2 is worth revisiting after finishing the book. Once Chapters 4, 6, 7, 8, 12, 13, and 16 have been covered, a reader should see how each of its steps would be implemented in practice.

1.8 Suggested Exercises

1.1 Define the difference between risk and uncertainty in the context of financial decision-making.

1.2 Explain why the time value of money matters for investment decisions.

1.3 List three types of financial markets and describe the assets traded in each.

1.4 Give one example of a financial problem where AI could be useful and one where classical statistical methods might be sufficient.

1.5 Write a short paragraph explaining why financial models should still be considered when using machine learning in finance.

1.6 Describe one financial institution and explain how it connects savers, borrowers, or investors.

1.7 Discuss why a model that performs well in a generic prediction task may still fail in a financial setting.

1.8 A gamble offers a 60% chance of gaining \$8,000 and a 40% chance of losing \$5,000. Compute the expected value, and explain, using the concept of a certainty equivalent, why a risk-averse investor might still reject it even though the expected value is positive.

1.9 Pick two of the challenges listed in Section 1.6.1 and describe, for each, a concrete financial scenario not used in the chapter in which it could cause an AI-based system to fail.

1.10 Using Table 1.7, compute the future value of \$5,000 invested for 20 years at 6%, and verify your answer is consistent with scaling the \$1,000 figure in the table.

1.11 A disease-screening test correctly identifies 90% of people who have a rare disease affecting 0.5% of the population, and has a 5% false-positive rate. Using Bayes' theorem as in Section 1.5.4, compute the probability that a person who tests positive actually has the disease, and explain why the result is counterintuitive.

1.12 Choose one entry from Table 1.8 and explain, in a short paragraph, why its financial problem could plausibly be framed as more than one of the table's task types (regression, classification, optimization, and so on).

1.13 Pick one milestone from Table 1.1 and explain, using material from later in this chapter, why the newer methods that followed it did not make it obsolete.

1.14 Continuing the sequential Bayesian updating example in Section 1.5.4, suppose a third, independent signal arrives: the transaction amount is a round number (e.g., exactly \$500), a pattern present in 40% of fraud cases but only 15% of legitimate ones. Using $\Pr(\text{Fraud} \mid \text{Flag}, \text{Location}) \approx 0.657$ as the new prior, compute the posterior probability of fraud after all three signals, and comment on how quickly it approaches certainty as independent evidence accumulates.

1.15 Using the Kelly criterion, compute the optimal betting fraction f^* for an even-money bet with a 70% win probability, and compute the long-run growth rate $g(f)$ at f^* and at $f = 0.9$, commenting on whether betting nearly everything is ever justified even when the win probability is this favorable.

1.16 A firm is deciding whether to enter a new market at a cost of \$3 million. Entry succeeds with probability 0.3, returning a net gain of \$12 million, or fails with probability 0.7, losing the full \$3 million. Compute the expected value of entering, the expected value under perfect information, and the resulting EVPI, and explain what the firm should conclude if a market-research report claiming near-perfect accuracy costs \$4 million.

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1.9 Further Reading

The following texts are useful starting points for the financial ideas introduced in this chapter:

- Brealey et al. (2025), *Principles of Corporate Finance*. A comprehensive introduction to corporate financial decision-making, including the time-value-of-money and valuation ideas this chapter introduces at a conceptual level and Chapter 5 develops formally.
- Hull (2022), *Options, Futures, and Other Derivatives*. The standard reference for derivatives pricing; useful background for the brief history of Black-Scholes-Merton option pricing sketched in this chapter's timeline and covered in full in Chapter 5.
- Bodie et al. (2024), *Investments*. Covers portfolio theory, market efficiency, and the risk-return trade-off introduced qualitatively in this chapter, previewing the quantitative treatment in Chapter 12.
- Campbell et al. (1997), *The Econometrics of Financial Markets*. A rigorous treatment of the statistical methods used to test market efficiency and analyze financial time series, going well beyond this chapter's introductory discussion.
- Goodfellow et al. (2016), *Deep Learning*. The standard reference for the deep learning methods behind the large language model and unstructured-data applications previewed in this chapter's history section and covered in Chapters 16 and 17.
- Chan (2017), *Machine Trading*. A practitioner-oriented introduction to building and testing systematic trading strategies, extending the case vignette in this chapter into the algorithmic trading material of Chapters 13 and 14.

These references cover foundational finance, asset pricing, and the use of quantitative methods in financial applications.

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